[EBOOK]

Bifrost: The Enterprise AI Gateway

A buyer's guide to deploying, governing, and scaling LLM infrastructure with the fastest enterprise AI gateway.

SOC 2 | GDPR | HIPAA | ISO 27001

github.com/maximhq/bifrost · docs.getbifrost.ai

$Gateway_Overhead

11µs

at 5,000 RPS

$P99_Latency

54×

faster than LiteLLM

$Throughput

9.5×

vs LiteLLM

$Providers

20+

1,000+ models

[OVERVIEW]

What Is Bifrost?

Bifrost is a high-performance, enterprise AI gateway written in Go, purpose-built for reliability, governance, and scale that unifies 20+ providers through a single OpenAI-compatible API.

Built in Go · Open source (Apache 2.0) · Only 11µs gateway overhead at 5K RPS

RELIABILITY
Automatic Failover

ROUTING
Adaptive Load Balancing

PERFORMANCE
Semantic Caching

ACCESS CONTROL
Virtual Key Governance

PROTOCOL
MCP Gateway

OBSERVABILITY
OpenTelemetry Built-In

SECURITY
Guardrails & RBAC

DEPLOYMENT
In-VPC Deployable

INFRASTRUCTURE
Single Binary · No Redis

INTELLIGENCE
Rule-Based Routing

[HOW IT SITS IN YOUR STACK]

Your Application
OpenAI / Anthropic / Bedrock SDK

↓

BIFROST GATEWAY
Go · ~11µs Overhead at 5K RPS · OpenAI API

Intelligent Routing
Routing + Fallback + Adaptive Load Balancing

↓

AI Providers
OpenAI · Anthropic · Google · AWS Bedrock · Azure · Mistral · Cohere · Groq · Vertex AI · + more

[BENCHMARKS · BIFROST VS LITELLM]

Performance Numbers That Matter

AWS t3.medium · 2 vCPU · 4 GB RAM · 500 RPS · 60-second sustained load · OpenAI Tier 5

$Throughput

9.5×

424 vs 44.84 req/s

$P99_Latency

54×

1.68s vs 90.72s

$Success_Rate

100%

vs 88.78% LiteLLM

$Memory

68%

120 MB vs 372 MB

[GATEWAY OVERHEAD · 60S MOCK TEST]

$Bifrost (Go)

11µs

$LiteLLM (Python)

40ms

40× less overhead

METRIC	BIFROST	LITELLM	DELTA
P50 Latency	804ms	38.65s	48× faster
Max Latency	6.13s	92.67s	15× faster
Peak Memory	120 MB	372 MB	68% less

[ARCHITECTURE]

Why Go Beats Python for AI Gateways

[PYTHON CHALLENGE]

Global Interpreter Lock

Python's GIL prevents true parallelism, forcing the interpreter to execute one thread at a time. Under high concurrency, this creates a bottleneck.

Async Overhead

Python's asyncio adds overhead in context switching and event loop management, especially with thousands of concurrent requests.

Memory Management

Python's dynamic typing and garbage collection consume more memory and can introduce latency spikes.

External Dependencies

Production Python deployments often require Redis for caching and rate limiting, adding operational complexity.

[BIFROST GO ADVANTAGE]

Native Goroutines

Go's goroutines enable handling thousands of concurrent requests with minimal memory overhead. No GIL, no bottlenecks.

Compiled Performance

As a compiled language, Go eliminates interpretation overhead and provides predictable, low-latency execution.

Memory Efficiency

Connection pooling with efficient memory reuse and lightweight goroutines reduce RAM consumption.

Built-in State Management

Bifrost handles configuration, logging, and state management internally without requiring external databases.

Docker Size: Bifrost ~80 MB · LiteLLM 700 MB+ · Setup: <30 seconds vs 2-10 minutes · Language: Go (compiled) vs Python (interpreted)

[OPEN SOURCE FEATURES]

What You Get Out of the Box

OSS tier · self-hosted · zero external dependencies

Drop-in Replacement

Replace your existing SDK with just one line change. Compatible with OpenAI, Anthropic, LiteLLM, Google Genai, Langchain and more.

Automatic Fallbacks

Automatic failover between providers ensures 99.999% uptime for your applications.

Virtual Key Governance

Create different virtual keys for different use-cases with independent budgets and access control.

Semantic Caching

Intelligent response caching based on semantic similarity. Reduce costs and latency by serving cached responses for semantically similar requests.

Built-in Observability

Out-of-the-box OpenTelemetry support for observability. Built-in dashboard for quick glances without any complex setup.

MCP Gateway

Centralize all MCP tool connections, governance, security, and auth. Your AI can safely use MCP tools with centralized policy enforcement.

[MCP GATEWAY]

MCP Gateway: Governed Tool Execution for AI Agents

Bifrost acts as both MCP client (connecting to external tool servers) and MCP server (exposing tools to clients like Claude Desktop) in a single deployment.

$Overhead	$Token_Savings	$Latency_Drop	$Providers
11µs	50%+	40%	20+

[CAPABILITIES]

- ✓ STDIO, HTTP, and SSE transports
- ✓ OAuth 2.0 with auto token refresh + PKCE
- ✓ Explicit tool execution with full control over approval and conversation flow
- ✓ Agent Mode with configurable auto-approval
- ✓ Tool filtering per virtual key (strict allow-lists)
- ✓ Federated auth - existing enterprise APIs as MCP tools
- ✓ Code mode reduces token usage by 50%+ when using multiple MCP servers

[SINGLE DEPLOYMENT · DUAL ROLE]

MCP CLIENT

Connects to external tool servers.

Auto-discovers capabilities at startup.

Filesystem · Databases · APIs · Custom

↓

MCP SERVER

Exposes tools via single gateway URL.

Claude Desktop · Cursor · Any MCP client.

Unified discovery + centralized security

[MCP · CODE MODE]

Code Mode: Token Efficiency for Multi-Tool Orchestration

Classic MCP sends all tool schemas on every request which is expensive at 3+ servers. Code Mode lets AI write Python to orchestrate tools in one round-trip.

DIMENSION	CLASSIC MCP	BIFROST CODE MODE
Token overhead	All schemas sent every request	**50%+ token reduction**
Execution latency	N tools = N sequential calls	**1 round-trip per workflow**
Multi-tool orchestration	Sequential only	**Python orchestrates in one pass**
Scalability	Degrades at 3+ servers	**Scales to any number**
Error handling	LLM retries each tool call	**Python try/catch in sandbox**
Latency improvement	Baseline	**40% lower latency**

[GOVERNANCE]

Virtual Keys: The Governance Primitive That Controls Everything

Every consumer – team, project, or application gets its own key with independent controls. Virtual keys are the primary governance entity in Bifrost.

Budget & Rate Limits

Dollar limits + request/token rate limits at virtual key, team, and customer levels. Per-provider throttling. Flexible reset windows.

Model & Provider Routing

Route specific keys to specific models or providers. Weighted load balancing. A/B test models via config, not code deployments.

MCP Tool Filtering

Strict allow-lists control which MCP tools a virtual key can access. Dev keys can't call production tools by accident.

Access Control

Model-specific restrictions, provider exclusions, custom permission scopes. Integrates with Okta and Entra via OIDC.

[GOVERNANCE HIERARCHY]

Organization
Global limits & compliance policies

↓

Team
Team-level budgets & model access

↓

Virtual Key
Per-consumer controls & tool access

↓

Request
Per-request routing & guardrails

[FEATURE · OPENTELEMETRY]

Native OTLP Integration for Distributed Tracing

OTLP protocol support for traces and metrics. Connect to Grafana Cloud, Datadog, New Relic, Honeycomb, or self-hosted collectors. All traces follow OpenTelemetry semantic conventions.

[PROTOCOL SUPPORT]

HTTP (OTLP/HTTP)
Port 4318
REST API for traces and metrics

gRPC (OTLP/gRPC)
Port 4317
High-performance binary protocol

[SPAN ATTRIBUTES]

- ✓ Request parameters (model, temp, tokens)
- ✓ Input messages and output responses
- ✓ Performance metrics (latency, tokens)
- ✓ Provider and model information
- ✓ Virtual key and user context
- ✓ Error details and stack traces
- ✓ Cost tracking per request
- ✓ Tool calls and function execution

[OBSERVABILITY]

Built In Observability

Every Bifrost deployment includes Prometheus metrics, OpenTelemetry tracing, and a real-time web dashboard, zero config, zero sidecars.

Prometheus Metrics

Native /metrics endpoint. HTTP-level and upstream provider metrics: requests, latency, success rates, token counts, provider health.

OpenTelemetry Tracing

OTLP integration. Send traces to Grafana, New Relic, Honeycomb. Just point and go, no instrumentation code needed.

Real-Time Web Dashboard

Monitor spend per key, model, and team from the built-in web UI. No external tools, no separate deployment or subscription.

Datadog Connector (Enterprise)

Native Datadog integration for APM traces, LLM Observability, and custom metrics. One config option, full trace pipeline.

Log Exports (Enterprise)

Automated export of request logs and telemetry to S3, GCS, Azure Blob, or custom data lakes for long-term retention.

Audit Logs (Enterprise)

Immutable audit trails per request/key/model/response. Satisfies SOC 2, GDPR, HIPAA, ISO 27001 audit evidence requirements.

[ENTERPRISE FEATURES]

Enterprise-Grade Controls

Production reliability, security, and compliance, built into the gateway.

Adaptive Load Balancing

Automatically optimizes traffic distribution across provider keys and models based on real-time performance metrics.

Clustering (Redis-backed)

High availability deployment with automatic failover and load balancing. Peer-to-peer clustering where every instance is equal.

Guardrails

Automatically detect and block unsafe model outputs with real-time policy enforcement and content moderation across all agents.

Governance (Okta · Entra)

SAML support for SSO and Role-based access control and policy enforcement for team collaboration.

Vault Support (HashiCorp · AWS)

Secure API key management with HashiCorp Vault, AWS Secrets Manager, Google Secret Manager, and Azure Key Vault integration. Store and retrieve sensitive credentials using enterprise-grade secret management.

Audit Logs (SOC 2 · GDPR · HIPAA)

Comprehensive logging and audit trails for compliance and debugging.

[ENTERPRISE · ADAPTIVE LOAD BALANCING]

Adaptive Load Balancing: Performance-Based Routing

Automatically distributes traffic based on success rates, latency patterns, and provider health. Two-level optimization: provider selection (best provider for a model) + key selection (best API key within that provider).

[HOW IT WORKS]

1. **Collect Metrics**
   Track success rates, latency, errors per provider/key
2. **Calculate Scores**
   Health scores based on recent performance history
3. **Provider Selection**
   Choose best provider for model based on scores
4. **Key Selection**
   Within provider, select best API key independently
5. **Continuous Adaptation**
   Scores update real-time, routing adjusts automatically

[KEY BENEFITS]

- ✓ Automatic failover on provider degradation
- ✓ Automatically detect provider TPM limits
- ✓ Slope detection to mark keys recovering or degrading
- ✓ Optimal key selection within each provider
- ✓ Circuit breaker integration
- ✓ Dynamic weight adjustment according to changing traffic patterns
- ✓ Cross node synchronization
- ✓ Minimal configuration overhead
- ✓ Real-time performance monitoring
- ✓ Predictive scaling capabilities

[ENTERPRISE · GUARDRAILS]

Guardrails: Content Safety on Every Request

Apply guardrails before or after LLM calls. Block harmful inputs, redact PII in outputs, enforce custom policies per workspace, tenant, or API key with full audit evidence for compliance.

Input Controls

- Stop bad inputs early
- Fewer incidents, safer launches
- Prompt attack detection
- PII & sensitive data blocking
- Jailbreak prevention

Output Safety

- Sanitize model outputs
- Selective redaction
- Policy-guided rewrites
- Protected material checks
- Auto-block or allow with override

Governance & Audit

- Per workspace / tenant / API key
- Process live or batch (cost-efficient)
- Targeted governance policies
- Compliance-ready audit evidence
- Decisions & overrides logged

[ENTERPRISE · CLUSTERING]

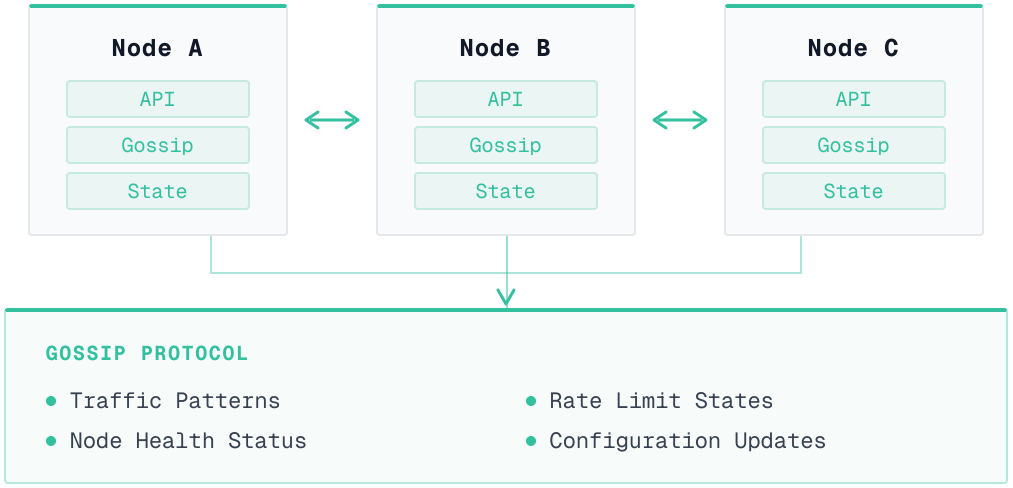
Cluster Mode: High-Availability with Automatic Service Discovery

Gossip-based P2P synchronization across nodes. Automatic service discovery, zero-downtime deployments, distributed state management without external coordinators like ZooKeeper.

[CLUSTERING CAPABILITIES]

- **Automatic Service Discovery**
  New nodes join cluster automatically via K8s, Consul, etcd
- **Gossip-Based Sync**
  State propagates via SWIM protocol - traffic patterns, rate limit states, node health, configuration changes, model performance, and provider weights
- **Zero-Downtime Deployments**
  Rolling updates with health checks, graceful shutdown
- **Distributed State**
  Config, budgets, rate limits sync across all nodes
- **No External Coordinator**
  Self-contained clustering, no ZooKeeper/etcd required

[CLUSTER TOPOLOGY]



[ENTERPRISE · RBAC & IDENTITY]

Role-Based Access Control with SSO Integration

OpenID Connect integration with Okta and Microsoft Entra ID for single sign-on. Users auto-provisioned on first login with roles and team memberships synced from identity provider.

[ROLE HIERARCHY]

Admin
Full access: users, keys, config, audit

Developer
Create/manage virtual keys, view logs

Viewer
Read-only access to logs and analytics

[USER-LEVEL BUDGETING]

`Customer` → `Team` → `User` → `Virtual Key`

[IDENTITY PROVIDERS]

- ✓ **Okta** - Full OIDC integration
- ✓ **Microsoft Entra ID** - Azure AD SSO
- ✓ **Google OAuth** - OSS tier
- ✓ **GitHub OAuth** - OSS tier
- ✓ Automatic user provisioning
- ✓ Role sync from identity provider
- ✓ Team membership sync
- ✓ Group-based access control

[ENTERPRISE · AUDIT LOGS]

Audit Logs: Immutable Audit Trails for Compliance

SOC 2, GDPR, HIPAA, and ISO 27001 compliance. Full request provenance: key → model → response. Automated compliance monitoring, policy enforcement, and audit trail generation.

[AUDIT TRAIL CAPTURES]

- **User Actions**
  Login, logout, key creation, config changes
- **Request Metadata**
  Virtual key, user, team, customer, timestamp
- **Model Usage**
  Provider, model, tokens, cost per request
- **Access Control**
  Authorization decisions, role assignments
- **Compliance Events**
  Policy violations, guardrail triggers

[COMPLIANCE REPORTS]

- SOC 2 Type II audit reports
- GDPR data access requests
- HIPAA compliance evidence
- ISO 27001 audit trails
- User activity summaries
- Budget utilization reports
- Model usage breakdowns
- Security incident logs

[ENTERPRISE · PROMPT REPOSITORY]

Prompt Repository: Version Control for Your AI Prompts

Centrally manage, version, and deploy prompts across your AI stack. Attach multiple deployment strategies per prompt version. Dynamically update prompts without code changes. Git-based 2-way sync supported.

[CORE FEATURES]

- **Built-in prompt versioning**
  Track every version with diffs and metadata
- **Full-fledged prompt playground**
  Test, iterate, and evaluate before deploying
- **Sync with version control**
  Connect to Git repos; enable 2-way sync
- **Targeted governance**
  Different policies per user, customer, provider, model, or virtual key
- **Dynamic prompt updates**
  Change prompts without code deployments

[DEPLOYMENT STRATEGIES]

- `gpt-4o` → use v3 of customer support prompt
- `claude-opus-4-5` → use v8 of same prompt
- `customer xyz` → use gpt-4o + v5 of prompt
- `env=beta` → route 40% traffic to v3
- A/B test prompt versions via config, not code
- Per-provider prompt override rules
- Per-customer prompt customization
- Gradual rollout with traffic splitting

[ENTERPRISE · MAXIM INTEGRATION]

maxim

End-to-end AI engineering workflows via UI or SDKs

Python · JS · TS · GO · Java

CI/CD

- GitHub Actions
- Jenkins
- circleci
- GitLab

FRAMEWORKS

- LangChain
- LangGraph
- crewai
- LiveKit
- VAPI
- n8n

MODEL PROVIDERS

- OpenAI
- ANTHROP\C
- Google
- Mistral AI
- groq
- together.ai
- Fireworks AI

DATA WAREHOUSE

- snowflake
- Google BigQuery

CLOUD OBSERVABILITY

- DATADOG
- new relic
- Grafana

+ any OTel collectors

+ many more

[ENTERPRISE · LOG EXPORTS]

Log Exports

Export LLM request logs and telemetry to any storage destination. Supports compliance archival, analytics pipelines, and real-time streaming across all major cloud providers.

Scheduled Exports

- Daily, weekly, or monthly automated exports
- Configurable export schedules
- Backfill and on-demand exports
- Real-time streaming mode

Destinations & Formats

- Object stores, data warehouses, data lakes
- JSON, CSV, Parquet, custom formats
- S3, GCS, Azure Blob, BigQuery, Redshift
- Custom webhook endpoints

Filtering & Compliance

- Export specific data subsets
- Custom field transformations
- Meet data retention requirements
- SOC 2, GDPR, HIPAA audit compliance

[ENTERPRISE · CUSTOM PLUGINS]

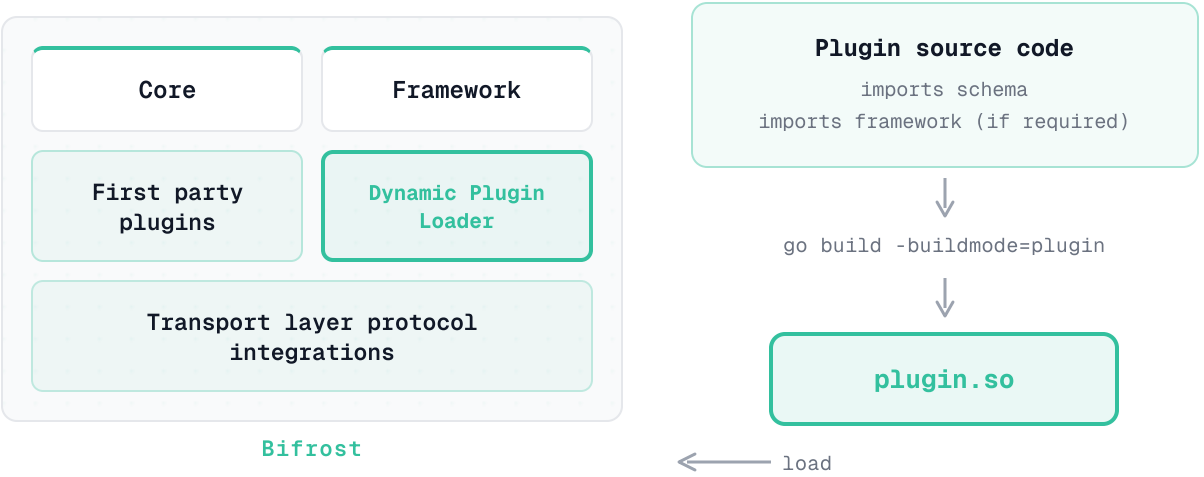
Custom Plugins: Extensible Middleware for Custom Logic

Build Go or WASM plugins to extend Bifrost without forking. Plugins execute at request/response boundaries, perfect for custom auth, transforms, mocking, or business logic.

[PLUGIN USE CASES]

- ✓ Custom authentication schemes
- ✓ Request/response transformation
- ✓ Mock AI responses for testing
- ✓ Audit logging to external systems
- ✓ Custom cost tracking & billing
- ✓ PII redaction & data masking
- ✓ Regional routing by IP address
- ✓ Dynamic prompt injection
- ✓ Rate limiting extensions

[ARCHITECTURE]



[ENTERPRISE · AUDIT & COMPLIANCE]

Compliance, Audit Reports & Identity

Comprehensive audit trails, immutable logs, and compliance reports for SOC 2, GDPR, HIPAA, and ISO 27001. Integrated with enterprise identity providers via SAML 2.0 and OpenID Connect for unified access management.

[IDENTITY & ACCESS]

- **SAML 2.0 & OpenID Connect**
  SSO integration with Okta, Entra ID, Google
- **Active Directory & LDAP sync**
  User auto-provisioning on first login
- **Role sync from identity provider**
  Group-based access control (GBAC)
- **User-level governance**
  Individual authentication + budget allocation per user
- **SCIM provisioning**
  Automated user lifecycle management

[COMPLIANCE REPORTS]

- SOC 2 Type II audit reports
- GDPR data access & erasure requests
- HIPAA compliance evidence packages
- ISO 27001 audit trail exports
- User activity summaries
- Budget utilization reports
- Model usage breakdowns
- Security incident logs

[COMPARISON]

Bifrost vs. The Alternatives

FEATURE	BIFROST	LITELLM	HAPROXY	ENVOY AI GW
Language / Runtime	**Go (compiled)**	Python (async)	C	C++ (Envoy) + Go
Gateway overhead	**11 µs**	40 ms	<1 ms (raw proxy)	~1-5 ms
Object Pooling	✓	✗	✗	✗
Semantic Caching	✓	✗	✗	✗
Adaptive Load Balancing	✓	✗	✗	✗
Geo-Aware Routing	✓	✗	✓	✗
Backpressure Handling	✓	✗	✓	✓
MCO Code Mode	✓	✗	✗	✗
P2P Tool Hosting	✓	✗	✗	✗
MCP Agent Mode	✓	✓	✗	✗
Zero-Config Startup	✓	✗	✗	✓
Prompt Studio	✓	✗	✗	✗
A2A Gateway	✓	✗	✗	✗
Mock Responses	✓	✗	✗	✗

[QUICK START]

Up and Running in Under 30 Seconds

No config files · No Redis · No databases · Single binary

01 Install

```
# Option 1: npx (fastest)
$ npx -y @maximhq/bifrost

# Option 2: Docker
$ docker run -p 8080:8080 maximhq/bifrost

# Option 3: Go SDK
$ go get
github.com/maximhq/bifrost/core@latest
```

One command. No config files, no Redis, no databases required.

02 Configure via Web UI

```
# Open the dashboard
$ open http://localhost:8080

# Add API keys for providers
# Configure models and weights
# Set up fallback chains
```

Browser-based dashboard. Add keys, set fallbacks. No YAML files.

03 Swap one line of code

```
# Before: direct to OpenAI
base_url="https://api.openai.com"

# After: through Bifrost
base_url="http://localhost:8080/openai"

# Everything else stays the same
```

Works with OpenAI, Anthropic, LiteLLM, LangChain, Vercel AI SDK.

Compatible: OpenAI SDK · Anthropic SDK · Bedrock SDK · GenAI SDK · LiteLLM · LangChain · PydanticAI · Vercel AI SDK

[DEPLOYMENT]

Flexible Deployment: Your Infrastructure, Your Rules

Self-Hosted

- npx
- Docker
- Binary

Single binary, zero dependencies. Deploy anywhere. Full control over data, config, and networking.

Kubernetes / Helm

- K8s
- Helm charts
- Horizontal scaling

Official Helm charts in the repo. Horizontal scaling, readiness probes, and rolling deployments out of the box.

In-VPC (Enterprise)

- AWS VPC
- GCP VPC
- Azure VNet

Deploy within your private cloud with VPC isolation. Data never leaves your network boundary.

Multi-Cloud

- AWS
- GCP
- Azure
- Cloudflare

Route to AWS Bedrock, Google Vertex, Azure OpenAI, and Cloudflare Workers AI simultaneously.

✓ SOC 2 Type II ✓ GDPR ✓ HIPAA ✓ ISO 27001

Build AI Infrastructure That Never Goes Down.

Bifrost unifies 20+ providers behind a single OpenAI-compatible API with automatic failover, virtual key governance, native MCP support, and enterprise-grade observability.

BOOK A DEMO | ENTERPRISE OFFERING | DOCS

✓ 2.3k GitHub stars ✓ Discord community

SOC 2 | GDPR | HIPAA | ISO 27001